Placement Portal Application

Course: MAD-1 Project

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Problem Statement:

Institutes require an efficient system to manage campus recruitment processes involving students, companies, and placement drives. Traditional systems such as spreadsheets and manual record keeping make it difficult to track student applications, approve companies, and manage recruitment drives.

This project implements a **Placement Portal Application** that digitizes the placement process and provides role-based access for Admin, Company, and Student users.

The system allows:

- Admin to manage companies and placement drives
- Companies to create recruitment drives
- Students to apply for jobs and track their application status

The goal is to **automate placement workflow**, prevent duplicate applications, and maintain a structured database of recruitment activities.

Approach to the Problem:

The application was designed as a **Flask based web system with role-based authentication**.

The development process followed these steps:

1. **Requirement Analysis**
    - Identified system roles: Admin, Company, Student
    - Defined functionalities for each role.
2. **Database Design**
    - Designed relational tables using SQLite.
    - Relationships were established between students, companies, placement drives, and applications.

3. **Backend Development**

    - Flask was used to implement routes and business logic.
    - Role-based access was implemented using session authentication.

4. **Frontend Development**

    - Jinja2 templates were used to dynamically render pages.
    - Bootstrap was used for responsive UI design.

5. **Testing**

    - Prevented duplicate student applications
    - Ensured only approved companies can create drives
    - Verified correct role-based permissions.

Frameworks and Libraries Used:

	Technology	Purpose
1.)	Flask	Backend web framework
2.)	SQLite	Database management
3.)	Jinja2	HTML templating
4.)	Bootstrap	Responsive UI design
5.)	HTML/CSS	Frontend structure and styling

System Roles and Functionalities:

1.) Admin

- Approve or reject company registrations
- Approve or reject placement drives

- View all students, companies, and drives
- Blacklist or deactivate accounts
- Search students or companies

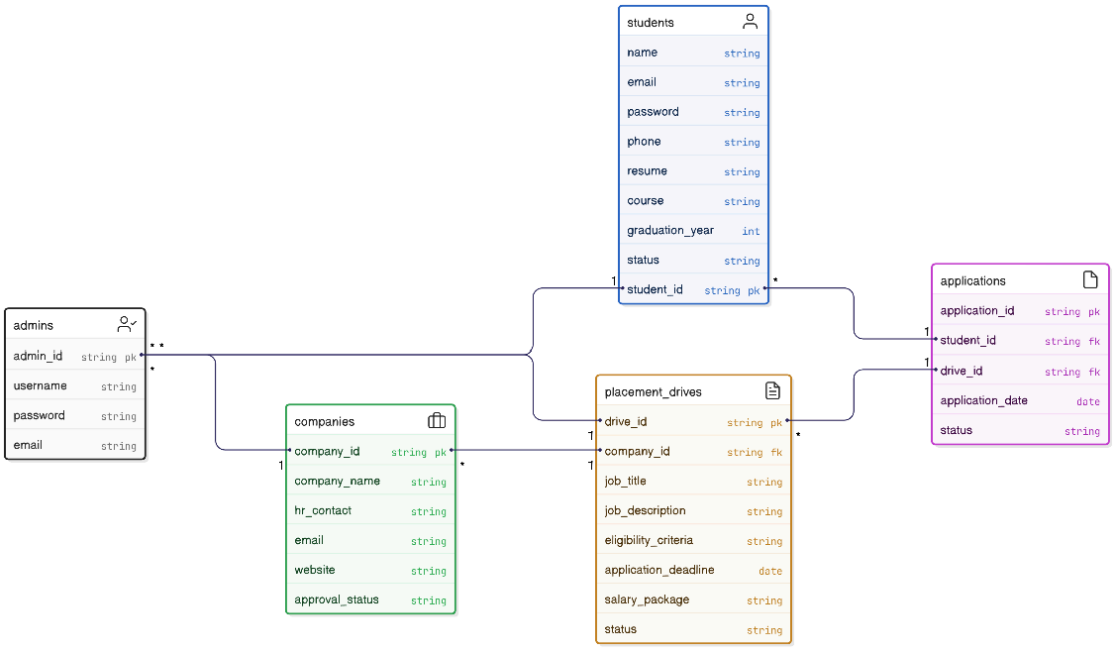
2.) Company

- Register and create company profile
- Create placement drives
- View student applications
- Update application status

3.) Student

- Register and update profile
- View approved placement drives
- Apply for drives
- Track application status and placement history

Database Design (ER Diagram):



AI / LLM Declaration:

Artificial Intelligence tools were used **only for UI design assistance**.

Specifically:

- AI tools were used to help design **HTML and CSS layout** for better visual presentation.
- All backend logic, database design, and business functionality were implemented manually.

No AI tools were used for generating application logic or database schema.

Conclusion

The Placement Portal Application successfully demonstrates a role-based recruitment management system using Flask and SQLite. The platform simplifies campus placement operations by enabling companies to create drives, students to apply for jobs, and administrators to manage recruitment activities efficiently.

The system prevents duplicate applications, ensures proper approval workflows, and maintains a complete record of placement history. The project highlights practical full-stack web development using Flask, Jinja2, Bootstrap, and SQLite.

Future improvements could include:

- Email notifications
- Placement analytics dashboards
- Resume filtering
- API integration for external job platforms